

COMP7035 – Python for Data Analytics and AI

Assignment 1: Programme Review from Messy Records

Background

At the end of the semester, the programme office must select students for a limited number of academic follow-up places. You are given four files: a score export, a student list, a weekly attendance log, and a policy file containing the score weights, review cutoff, and number of available places.

The files use different formats and contain incomplete or overlapping information. Your task is to inspect, clean, and combine these records into a reliable student-level dataset, then use the available evidence to support a recommendation for the programme coordinator.



From a messy export to a student list the programme office can use.

Getting started

The starter notebook `hw1_template.ipynb` is released together with the four data files `scores.csv`, `students.json`, `engagement.log`, and `policy.json`. Put the notebook and these four files in one folder, rename the notebook to `hw1_<ID>.ipynb`, enter your eight-digit student ID in the first code cell, and run the notebook from the top. Change only the `STUDENT_ID` value in that provided cell; your ID is used only to name your output files. Everyone works on the same data, so numeric results will agree across classmates – your method, your choices, and your report are what make the work yours. Use the Python standard library throughout.

What you need to do

You are given four data files containing overlapping student records. There is no predefined cleaning procedure. Your task is to **inspect the files, identify data-quality problems, and develop a reasonable and reproducible set of rules for combining them.**

Before consulting or using any AI tool, first work through the assignment independently. Do not rely on AI to perform the key analytical thinking or make critical judgments for you. Your cleaning rules, conflict resolutions, interpretation of missing evidence, recommendation logic, meaningful question, and additional experiment must originate from you, and you must be able to explain and defend them yourself.

First, explore the four files and **document the issues you find**, including duplicate records, conflicting values, missing values, and differences between data sources. **Explain which records or sources you consider reliable and how you resolve conflicts.** In particular, **distinguish between evidence that a student did not participate and cases where participation data are simply missing.**

Using these rules, **create `hw1_<ID>_clean.csv`**. The file should contain **exactly one row per student** and include the final accepted scores, available attendance or engagement evidence, and the weighted total score. The score weights, review cutoff, and number of available places **must be read from `policy.json`**.

Based on the cleaned records, **identify the students you recommend for the available places**. Your notebook should **show the records and evidence supporting these recommendations**, including any uncertainty or cases that may require human review. The recommendations should be treated as **decision support rather than automatic decisions**.

In addition to the required analysis, **investigate one meaningful question** that arises from your exploration of the data. **Explain why the question matters and answer it using code and evidence**.

You must also **perform one additional analytical experiment of your own**. This experiment should modify, challenge, or extend the way the evidence is used – for example, by testing an alternative treatment of uncertain records or by conducting a sensitivity analysis. **Compare the results before and after this change**, and explain whether it changes or strengthens your final recommendations.

Make the reasoning visible in the notebook, especially the judgment calls that a reviewer should inspect first. Use functions and meaningful variable names. The notebook **must run successfully from top to bottom in a folder that contains the notebook and the four provided data files**.

In addition to the notebook, **write a separate analysis report named hw1_<ID>_report.pdf**. The report should explain the main data-quality problems and cleaning rules, present and justify your recommendations, discuss uncertainty and cases requiring human review, summarise your meaningful question and additional experiment, and state the main limitations. It should be understandable without running the notebook and should use relevant tables, figures, or numerical evidence from your analysis. **Use A4 page size, Times New Roman, 12-point font, and single line spacing. The report must not exceed five pages.**

What to submit

Submit **three files** on Moodle: hw1_<ID>.ipynb, hw1_<ID>_clean.csv, and hw1_<ID>_report.pdf. Do not submit the four provided data files.